



angles?  $\theta = ?$

#### 4.4 Solving Trigonometric Equations

$x = ?$

What is a trig equation?

- A trig equation is an equation that contains one or more trigonometric functions.

How is it similar to a trig identity?

- A trig equation *can* be but *does not have to* be a trig identity
  - A trig identity is an equation that is true for **all values** of the variable for which expressions of both sides of the equation are defined
- A trig equation that is not an identity is only true for **certain values** of the variable.

What does it mean to **SOLVE** a trig equation?

- Much like solving a linear equation, we are looking for all of the values of the variable that makes the equation true.

#### General Solutions

i) If  $\cos(x) = \cos(\alpha)$ , then  $x = 2\pi k \pm \alpha$ ,  $k \in \mathbb{Z}$

Ex.1) Find the exact solutions for  $2\cos(x) + \sqrt{3} = 0$  for  $x \in [0, 2\pi]$ .

$$x = ?$$

$$\cos(x) = -\frac{\sqrt{3}}{2}$$

$$RAA = \cos^{-1}\left(\frac{\sqrt{3}}{2}\right)$$

$$RAA = \frac{\pi}{6}$$

$\cos$  is  $(-)$  in QII & QIII

$$\Rightarrow x_1 = \pi - RAA$$

$$= \pi - \frac{\pi}{6}$$

$$x_1 = \frac{5\pi}{6}$$

$$x_2 = \pi + RAA$$

$$= \pi + \frac{\pi}{6}$$

$$x_2 = \frac{7\pi}{6}$$

$$\therefore x \in \left\{ \frac{5\pi}{6}, \frac{7\pi}{6} \right\}$$

ii) If  $\sin(x) = \sin(\alpha)$ , then  $x = 2\pi k + \alpha$  or  $x = 2\pi k + (\pi - \alpha)$ ,  $k \in \mathbb{Z}$

Ex.2) Find the exact solutions for  $2\sin(2x) = \sqrt{3}$  for  $x \in [0, 2\pi]$ .

$$2x \in [0, 4\pi]$$

$$\sin(2x) = \frac{\sqrt{3}}{2}$$

$$\text{Let } A = 2x$$

$$RAA = \sin^{-1}\left(\frac{\sqrt{3}}{2}\right)$$

$$A = \frac{\pi}{3}$$

$\sin$  is  $(+)$  in QI & QII

$$A_1 = \frac{\pi}{3}$$

$$2x_1 = \frac{\pi}{3}$$

$$x_1 = \frac{\pi}{6}$$

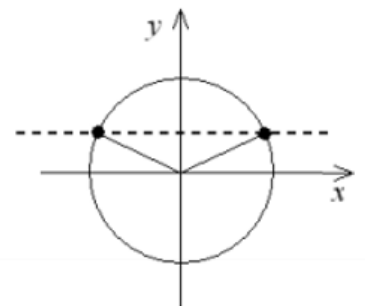
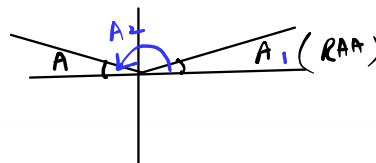
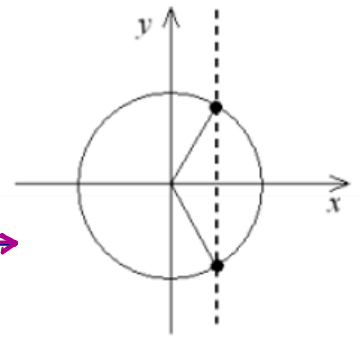
$$A_2 = \pi - RAA$$

$$2x_2 = \pi - \frac{\pi}{3}$$

$$x_2 = \frac{2\pi}{3}$$

$$x_2 = \frac{\pi}{3}$$

$$\therefore x \in \left\{ \frac{\pi}{6}, \frac{\pi}{3}, \frac{2\pi}{3}, \frac{5\pi}{6} \right\}$$



$$\boxed{x_1 = \frac{\pi}{6}}$$

$$\boxed{x_2 = \frac{\pi}{3}}$$

$$\therefore x \in \left\{ \frac{\pi}{6}, \frac{\pi}{3}, \frac{7\pi}{6}, \frac{4\pi}{3} \right\}$$

$$A_3 = 2\pi + \pi A_1$$

$$2x_3 = 2\pi + \frac{\pi}{3}$$

$$2x_3 = \frac{7\pi}{3}$$

$$\boxed{x_3 = \frac{7\pi}{6}}$$

$$A_4 = 2\pi + A_2$$

$$2x_4 = 2\pi + \frac{2\pi}{3}$$

$$2x_4 = \frac{4\pi}{3}$$

$$\boxed{x_4 = \frac{4\pi}{3}}$$

iii) If  $\tan(x) = \tan(\alpha)$ , then  $x = k\pi + \alpha$ ,  $k \in \mathbb{Z}$

Ex.3) Find the exact solutions for  $3\tan^2(\theta) = 1$  for  $\theta \in [0, 2\pi]$ .

$$\tan^2(\theta) = \frac{1}{3}$$

$$\tan(\theta) = \pm \frac{1}{\sqrt{3}}$$

$$R.A.A = \tan^{-1}\left(\frac{1}{\sqrt{3}}\right)$$

$$= \frac{\pi}{6}$$

$$\theta_1 = \frac{\pi}{6}$$

$$\theta_2 = \pi - \frac{\pi}{6}$$

$$\theta_3 = \frac{5\pi}{6}$$

$$\theta_3 = \pi + \frac{\pi}{6}$$

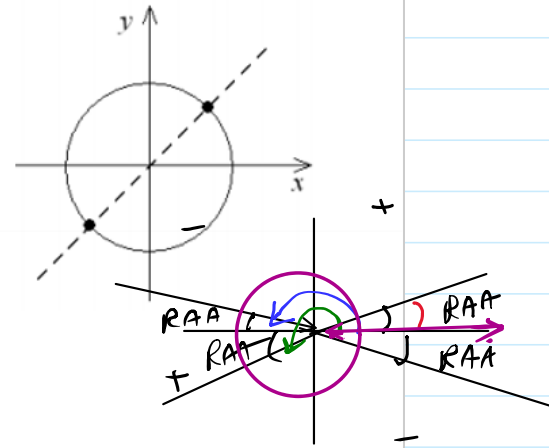
$$\theta_3 = \frac{7\pi}{6}$$

$$\theta_4 = 2\pi - R.A.A$$

$$= 2\pi - \frac{\pi}{6}$$

$$\theta_4 = \frac{11\pi}{6}$$

$$\therefore \theta \in \left\{ \frac{\pi}{6}, \frac{5\pi}{6}, \frac{7\pi}{6}, \frac{11\pi}{6} \right\}$$



Ex.4) Find the exact solutions for  $\tan^2(\theta) + 3\sec(\theta) + 3 = 0$ ,  $0 \leq \theta \leq 2\pi$

$$\sec^2\theta - 1 + 3\sec\theta + 3 = 0$$

$$\sec^2(\theta) + 3\sec(\theta) + 2 = 0$$

$$\text{Let } A = \sec(\theta)$$

$$\Rightarrow A^2 + 3A + 2 = 0$$

$$(A+2)(A+1) = 0$$

$$A + 2 = 0$$

$$A + 1 = 0$$

$$A = -2$$

$$\text{Sub. } A = \sec\theta$$

$$\Rightarrow \sec\theta = -2$$

$$\frac{1}{\cos\theta} = -2$$

$$\cos\theta = -\frac{1}{2}$$

$$R.A.A = \cos^{-1}\left(\frac{1}{2}\right)$$

$$= \pi$$

$$\therefore \tan^2\theta + 1 = \sec^2(\theta)$$

$$\tan^2(\theta) = \sec^2(\theta) - 1$$

$$= \frac{\pi}{3}$$

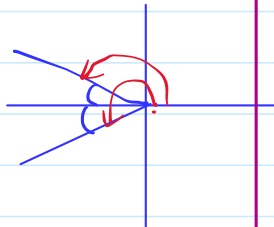
$\cos$  is  $(-)$  in  $\text{QII}$  &  $\text{QIII}$

$$\theta_1 = \pi - \frac{\pi}{3}$$

$$\boxed{\theta_1 = \frac{2\pi}{3}}$$

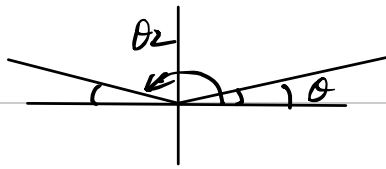
$$\theta_2 = \pi + \frac{\pi}{3}$$

$$\boxed{\theta_2 = \frac{4\pi}{3}}$$



$$\theta_3 = \pi$$

$$\therefore \theta \in \left\{ \frac{2\pi}{3}, \frac{4\pi}{3}, \pi \right\}$$



Ex.5) Find the solutions for  $3\cos^2(\theta) - \sin(\theta) - 1 = 0, 0 \leq \theta \leq 2\pi$

$$3[1 - \sin^2(\theta)] - \sin(\theta) - 1 = 0$$

$$3 - 3\sin^2(\theta) - \sin(\theta) - 1 = 0$$

$$-3\sin^2(\theta) - \sin(\theta) + 2 = 0$$

Let  $\sin(\theta) = A$

$$-3A^2 - A + 2 = 0$$

$$(3A - 2)(A + 1) = 0$$

$$3A - 2 = 0, \quad A + 1 = 0$$

$$A = \frac{2}{3}, \quad A = -1$$

$$\sin \theta = \frac{2}{3}, \quad \sin \theta = -1$$

$$RAA = \sin^{-1}\left(\frac{2}{3}\right)$$

$$\approx 0.73$$

$\sin$  is (+) in QI & QII

$$\theta_1 \approx 0.73$$

$$\theta_2 = \pi - 0.73$$

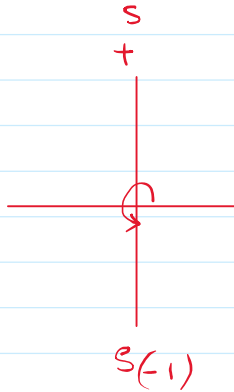
$$\theta_2 \approx 2.41$$

$$\cos^2 \theta = 1 - \sin^2 \theta$$

$$\sin \theta = -1$$

$$\theta_3 = \frac{3\pi}{2}$$

$$\therefore \theta \in \left\{ 0.73, 2.41, \frac{3\pi}{2} \right\}$$



Ex.6) Solve the equation  $\sin(4x) - \cos(2x) = 0, x \in [0, 2\pi]$

$$2\sin(2x)\cos(2x) - \cos(2x) = 0$$

$$\cos(2x)[2\sin(2x) - 1] = 0$$

$$\cos(2x) = 0$$

$$2\sin(2x) - 1 = 0$$

Let  $A = 2x$   
 $\cos(A) = 0$   
 $RAA = \frac{\pi}{2}$

$$A_1 = \frac{\pi}{2}$$

$$2x_1 = \frac{\pi}{2}$$

$$x_1 = \frac{\pi}{4}$$

$$A_2 = \frac{3\pi}{2}$$

$$2x_2 = \frac{3\pi}{2}$$

$$x_2 = \frac{3\pi}{4}$$

$$A_3 = \frac{5\pi}{2}$$

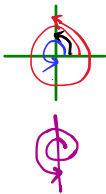
$$2x_3 = \frac{5\pi}{2}$$

$$x_3 = \frac{5\pi}{4}$$

$$A_4 = \frac{7\pi}{2}$$

$$2x_4 = \frac{7\pi}{2}$$

$$x_4 = \frac{7\pi}{4}$$



$$\sin(2x) = \frac{1}{2}$$

Let  $A = 2x$

$$RAA = A = \frac{\pi}{6}$$

$\sin$  is (+) in QI & QII

$$\Rightarrow A_1 = RAA$$

$$2x_1 = \frac{\pi}{6}$$

$$x_1 = \frac{\pi}{12}$$

$$A_2 = \pi - \frac{\pi}{6}$$

$$2x_2 = \frac{5\pi}{6}$$

$$x_2 = \frac{5\pi}{12}$$

$$A_3 = 2\pi + \frac{\pi}{6}$$

$$2x_3 = \frac{13\pi}{6}$$

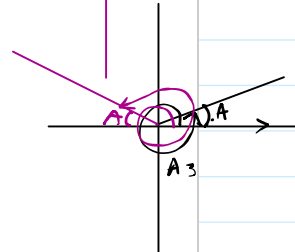
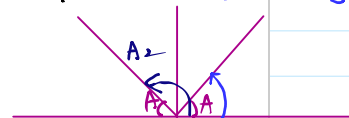
$$x_3 = \frac{13\pi}{12}$$

$$A_4 = 2\pi + \frac{5\pi}{6}$$

$$2x_4 = 2\pi + \frac{5\pi}{6}$$

$$2x_4 = \frac{17\pi}{6}$$

$$x_4 = \frac{17\pi}{12}$$



$$\theta \in \left\{ \frac{\pi}{4}, \frac{3\pi}{4}, \frac{5\pi}{4}, \frac{7\pi}{4}, \frac{\pi}{12}, \frac{5\pi}{12}, \frac{13\pi}{12}, \frac{17\pi}{12} \right\}$$

Ex.7) Solve  $\sin(2x) - \cos(2x) = 0, 0 \leq x \leq 2\pi$ 

$$\begin{aligned}\sin(2x) &= \cos(2x) \\ \frac{\sin(2x)}{\cos(2x)} &= \frac{\cos(2x)}{\cos(2x)} \\ \tan(2x) &= 1 (+)\end{aligned}$$

$$RAA = \tan^{-1}(1)$$

$$RAA = \frac{\pi}{4}$$

 $\tan \theta$  is (+) in QI & QIII

QI

$$\begin{aligned}2x_1 &= \frac{\pi}{4} \\ x_1 &= \frac{\pi}{8}\end{aligned}$$

Ex.8) Solve  $\tan(\theta)(\csc(\theta)+2)=0, 0 \leq \theta \leq 2\pi$ 

$$\tan(\theta) = \frac{\sin(\theta)}{\cos(\theta)} \Rightarrow \cos \theta \neq 0 \Rightarrow \theta \neq (2n+1)\frac{\pi}{2}; n \in \mathbb{Z}$$

$$\csc(\theta) = \frac{1}{\sin(\theta)} \Rightarrow \sin \theta \neq 0 \Rightarrow \theta \neq n\pi; n \in \mathbb{Z}$$

$$[\tan(\theta)][\csc(\theta) + 2] = 0$$

$$\tan \theta = 0, \quad \csc(\theta) + 2 = 0$$

$$\theta = \tan^{-1} 0,$$

$$\theta = 0$$

Unadmissible be  $\csc \theta = \frac{1}{\sin \theta}$

$$= \frac{1}{\sin 0}$$

$$= \frac{1}{0}$$

(Undefined)

$$\csc(\theta) = -2$$

$$\frac{1}{\sin(\theta)} = -2$$

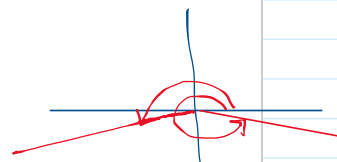
$$\sin \theta = -\frac{1}{2}$$

$$RAA = \sin^{-1}\left(-\frac{1}{2}\right)$$

$$= -\frac{\pi}{6}$$

$$\Rightarrow \theta_1 = \pi + RAA = \pi - \frac{\pi}{6}$$

$$\theta_1 = \frac{5\pi}{6}$$



$$\theta_2 = 2\pi - RAA$$

$$= 2\pi - \frac{\pi}{6}$$

$$\theta_2 = \frac{11\pi}{6}$$

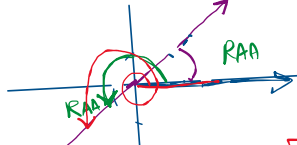
$$\therefore \theta \in \left\{ \frac{5\pi}{6}, \frac{11\pi}{6} \right\}$$

$$x \neq (2n+1)\frac{\pi}{2}$$

$$\cos x \because \cos\left((2n+1)\frac{\pi}{2}\right) = 0$$

$$\Rightarrow 2x \neq (2n+1)\frac{\pi}{2}; n \in \mathbb{Z}$$

$$\Rightarrow x \neq (2n+1)\frac{\pi}{4}$$



QIII

$$2x_3 = \pi + RAA$$

$$2x_3 = \pi + \frac{\pi}{4}$$

$$2x_3 = \frac{5\pi}{4}$$

$$x_3 = \frac{5\pi}{8}$$

$$2x_4 = 2\pi + 2x_3$$

$$2x_4 = 2\pi + \frac{5\pi}{4}$$

$$2x_4 = \frac{13\pi}{4}$$

$$x_4 = \frac{13\pi}{8}$$

OR

$$2x_4 = 3\pi + RAA$$

$$2x_4 = 3\pi + \frac{\pi}{4}$$

$$2x_4 = \frac{13\pi}{4}$$

$$x_4 = \frac{13\pi}{8}$$

$$x \in \left\{ \frac{\pi}{8}, \frac{9\pi}{8}, \frac{5\pi}{8}, \frac{13\pi}{8} \right\}$$